

CIS 106

Problems – Using Nested if and Compound Relational Conditions

For each problem, develop the IPO and Code.

1. The input to the problem is quantity of widgets. Your program should determine the price to charge based on the schedule below. Calculate the extended price (quantity x price). Calculate tax at 7%. Display the extended price, tax amount and total.

Quantity	Price
>10000	\$10
5000 to 10000	\$20
Below 5000	\$30

input	process	Output
Quantity of widgets	Nested if to determine unit price: - If quantity > 10000, price = \$10 - Else if 5000 <= quantity <= 10000, price = \$20 - Else, price = \$30	Extended price
	Calculate extended price = quantity * price	tax amount
	Calculate tax = extended price * 0.07	total cost
	Calculate total = extended price + tax	

2. Enter a part number and quantity Determine the cost per unit using the table below. Then calculate the total cost (quantity x unit cost). Display the part number, cost per unit and total cost. Note: Part number can be an integer but it can also be a string because you are not doing arithmetic on it. However in your code if statement be sure to compare using consistency, that is, if item == "10" when item is a string and if item == 10 when item is an integer.

Part	Unit Cost
10 <u>or</u> 55	1.00
99	2.00
80 <u>or</u> 70	3.00
All others	5.00

input	Process	output
Part number	Nested if to determine unit	Part number

	cost: - If part is "10" or "55", unit cost = \$1.00 - Else if part is "99", unit cost = \$2.00 - Else if part is "80" or "70", unit cost = \$3.00 - Else, unit cost = \$5.00	
quantity	Calculate total cost = quantity * unit cost	Part number
		total cost

3. Enter a principle amount of a CD and year to maturity of CD. Determine the interest rate based on the amount of the principle **and** maturity (see below). Calculate first year interest (principle x interest rate). Display principle, interest rate and the interest amount for first year.

Principle	Years to Maturity	Interest Rate
>\$100,000	5	6%
\$50,000 to \$100,000	10	5%
\$50,000 to \$100,000	5	4%
Any other principle and years		2%

input	process	output
Principle amount	Nested if to determine interest rate: - If principle > 100000 and years == 5, rate = 6% - Else if 50000 <= principle <= 100000: - If years == 10, rate = 5% - Else if years == 5, rate = 4% - Else, rate = 2%	Principle amount
years to maturity	Calculate interest = principle * rate	interest rate
		first year interest

4. Allow the user to enter number of concert tickets. The price per ticket depends on the volume (see below). Display the number of tickets, price per ticket and the total cost (number of tickets x Price Per Ticket).

Quantity Price Per Ticket

>=25 \$50
 10 to 24 \$60
 5 to 9 \$70
 Less 5 \$75

input	process	output
Number of tickets	Nested if to determine price per ticket: - If tickets >= 25, price = \$50 - Else if 10 <= tickets <= 24, price = \$60 - Else if 5 <= tickets <= 9, price = \$70 - Else, price = \$75	Number of tickets
	Calculate total cost = tickets * price per ticket	price per ticket
		total cost

5. The user will enter employee last name, salary and job level (as noted below). Use the job level to determine the bonus rate. Then compute bonus to be salary times bonus rate. Display employee last name and bonus.

Job Level Bonus Rate

10 and above 25%
 5 to 9 20%
 All others 10%

input	process	output
Lname	Nested if to determine bonus rate: - If level >= 10, bonus_rate = 25% - Else if 5 <= level <= 9, bonus_rate = 20% - Else, bonus_rate = 10%	Employee last name

Salary	Calculate bonus = salary * bonus rate	bonus amount
Job level		